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A Primer on Nonmarket Valuation

Second Edition

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Chapter 3
Collecting Nonmarket Valuation Data

Patricia A. Champ

Abstract This chapter describes how to collect primary and secondary data for nonmarket valuation studies. The bulk of this chapter offers guidance on how to design and implement a high-quality nonmarket valuation survey. Understanding the full data collection process will also help in evaluating the quality of data that someone else collected (i.e., secondary data). As there are not standard operating procedures for collecting nonmarket valuation, this chapter highlights the issues that should be considered in each step of the data collection process from sampling to questionnaire design to administering the questionnaire. While high-quality data that reflect individual preferences will not ensure the reliability or validity of estimated nonmarket values, quality data are a prerequisite for reliable and valid nonmarket measures.

Keywords Primary data · Secondary data · Nonmarket valuation survey · Sampling procedures · Study population · Sample frame · Survey mode · Survey documentation

The first two chapters in this book describe how individual choices are the foundation of nonmarket values. While that concept is relatively straightforward, one might wonder how we know about the choices an individual has made or would make. In other words, where do the data come from that reflect individuals' choices for nonmarket goods?

Sometimes market data can be used to infer choices about nonmarket goods. For example, I paid a premium price for the lot where I built my house. I did this because I liked that the lot backed up to open space. I could have purchased an almost identical lot in the same neighborhood that did not border the open space at a lower price. One could make an inference about my preference for open space from that market transaction. If there are many such transactions, one could use the data on lot sale prices in my neighborhood to infer the value of the open space behind my house.

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More often, choices over environmental goods are not observable in market transactions. For example, an individual could care deeply and be willing to pay for the preservation of an endangered tiger species. However, there might not be a tiger preservation program that provides an opportunity to do that. In that situation, the researcher must go directly to individuals and ask what choices they would make about a particular nonmarket good or service if they had a chance to make such choices. Direct questioning about choices is done in surveys.

Surveys are prevalent. The current survey-crazed environment makes it difficult for potential respondents to know whether a survey is legitimate or a marketing ploy. Therefore, good survey design and scientific practice are more important than ever.

The bulk of this chapter offers guidance on how to design and implement a high-quality nonmarket valuation survey. Understanding the full data collection process will also help in evaluating the quality of data that someone else collected (i.e., secondary data). While high-quality data that reflect individual preferences will not ensure the reliability or validity of estimated nonmarket values, and quality data are a prerequisite for reliable and valid nonmarket measures. As the saying goes, “garbage in, garbage out.” If data are not of sufficient quality, the corresponding nonmarket valuation estimates will be suspect.

While standard operating procedures do not exist for the collection of survey data, this chapter focuses on considerations and tradeoffs that are made in each step of the design and in administration of a nonmarket valuation survey. Throughout the chapter, the term “survey” refers to both the methods and the materials used to collect data. “Questionnaire” refers to the set of questions on the survey form. Following the discussion on developing and implementing a survey, the use of secondary data is discussed.

3.1 The Objective of the Nonmarket Valuation Study

Before worrying about data, the researcher must clarify the objectives of the study. The overall objective might be, for example, to inform policy, to assess the magnitude of damage to a natural resource, or to improve a nonmarket valuation method. Given the broad study objectives, the researcher must specify the economic question to be answered and decide whether equivalent surplus or compensating surplus will be measured. The economic question should be stated using the framework described in Chap. 2. If the study addresses a research question, the researcher should also develop testable hypotheses. The study objectives and economic questions will determine the most appropriate nonmarket valuation technique. In turn, the technique will determine what data are needed to estimate values.

In Chap. 1, Table 1.1 summarizes the broad steps in the valuation process. Data collection for nonmarket valuation falls under Step 5. The first order of business is to ascertain whether existing data can be used or whether it is necessary to collect new data. If new (i.e., primary) data need to be collected using a survey, decisions about the sampling techniques, mode of administration, and questionnaire content will be made in light of the overall study objectives and the specific economic questions to be addressed. If secondary data can be used to estimate nonmarket values, the researcher must decide where to get the data. As secondary data were once primary data, this chapter (in Sect. 3.2) largely focuses on conducting surveys to collect primary nonmarket valuation data. Section 3.3 then discusses secondary data sources.

3.2 Nonmarket Valuation Survey Methodology

The rapid changes in communication technology have had a huge impact on survey research. In the past, surveys were typically implemented start-to-finish using just one approach such as the telephone, the mail, or an in-person interviewer. Those days are gone.

It is now common for survey data to be collected using multiple approaches. For example, potential respondents can initially be contacted with a letter in the mail and given a choice of completing a paper or online questionnaire. A telephone call can then be used to remind nonrespondents to complete the questionnaire. Surveys implemented with more than one mode are referred to as mixed-mode surveys. As technology continues to change, new possibilities for collecting survey data will emerge. This chapter focuses on fundamental considerations that are likely to endure in the future for implementing a survey.

Before covering development and implementation of a nonmarket valuation survey, this section describes the process for deciding who will be invited to complete the survey. The survey objective will influence this decision. While some nonmarket surveys are conducted to provide values that can be generalized to the population of interest, there are other situations where the survey results are not intended to represent a broader population. For example, the economic experiments described in Chap. 10 may involve administering surveys to convenience samples drawn from student populations. The survey could be developed to allow for comparisons of values across similar groups as a test of economic theory or for tests of methodological issues related to nonmarket valuation. In both of these situations, the steps in development of the questionnaire and other survey materials will be similar. However, the situations will differ in the recruitment of survey respondents or in sampling procedures. In practice, the researcher makes decisions about sampling procedures and design of survey materials at the same time, as sampling options inform design and vice versa. However for the sake of exposition, survey sampling is described before survey design.

3.2.1 Sampling Procedures

There are two general approaches to sampling: probability sampling and nonprobability sampling (Fig. 3.1). Probability sampling involves a random selection process for inclusion of units in the sample where each unit of the study population has a known nonzero chance of being selected. In contrast, with nonprobability sampling, each individual in the population does not have a known nonzero probability of being

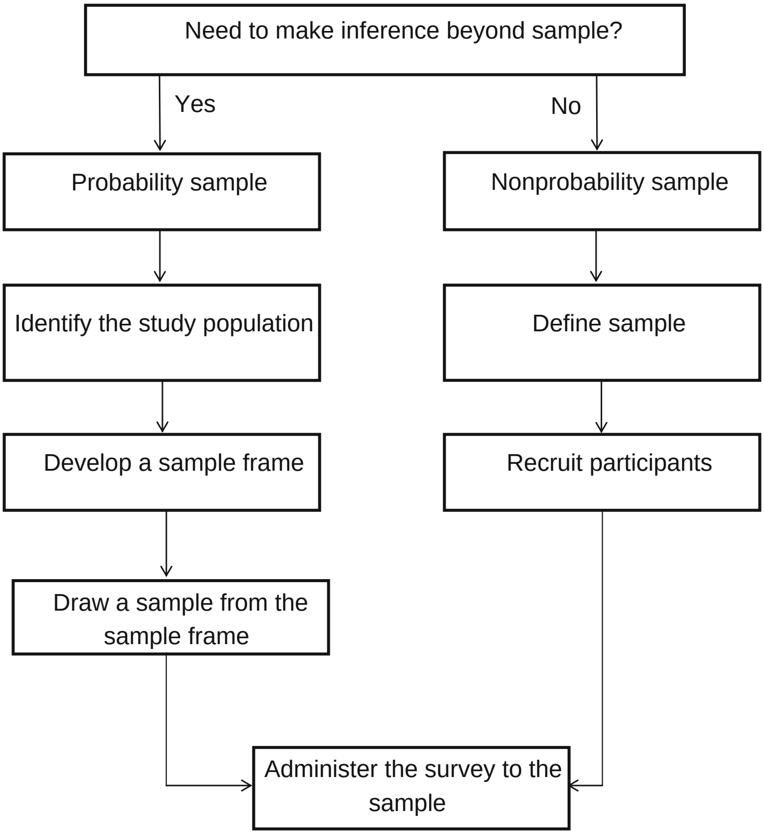

Fig. 3.1 Sampling

chosen for the sample. Examples of nonprobability sampling include using students in a class and recruiting subjects in a shopping mall. Nonprobability sampling is best-suited for studies that do not require generalization from the survey sample to a broader population. Such samples are frequently used to investigate methodological questions such as how responses are affected by survey design features.

The next three sections describe the steps associated with developing a probability-based sample. The sample design specifies the population of interest (study population), the sample frame, and the techniques for drawing a sample from the sample frame. Figure 3.1 outlines the steps in the sample design, while Fig. 3.2 identifies the potential sources of error that can arise during the survey process. Section 3.2.1.4 covers approaches for nonprobability sampling.

3.2.1.1 Study Population

Defining the study population begins with determining whose values are relevant. Consider a city interested in providing free Wi-Fi to a particular neighborhood. If the city would like to measure the benefits to neighborhood residents of providing

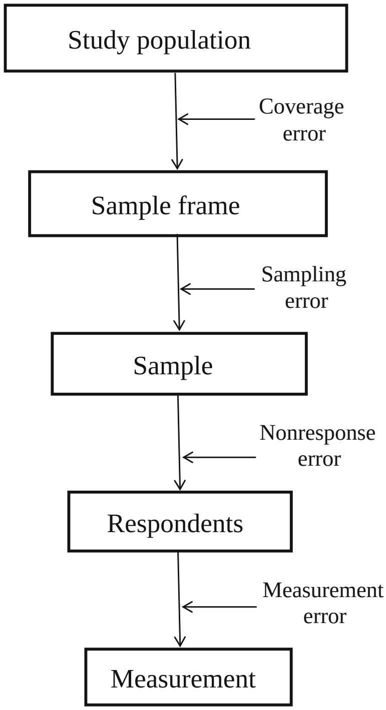

Fig. 3.2 Potential sources of error

free Wi-Fi, the appropriate study population is all residents who live within the neighborhood. This study population includes residents of the neighborhood who currently use Wi-Fi as well as those who do not. Including all neighborhood residents allows for measurement of the benefits to both current and potential Wi-Fi users.

However, the benefits of free Wi-Fi also can accrue to individuals who do not live in the neighborhood but visit the neighborhood for work or recreation. Further, it is possible that some individuals who neither live in nor visit the neighborhood could benefit from the provision of the free Wi-Fi. For example, the neighborhood might include school children who are economically disadvantaged, and some people beyond the neighborhood could benefit from knowing that the children have Internet access to help them with their schoolwork. This would be an example of passive-use values.

Identification of the relevant study population for measurement of passive-use values can be challenging because there is unlikely to be a clear geographical boundary beyond which passive-use value falls to zero or to some minimum level. And the boundary issue can be of great importance, as in valuing a good or program of national interest but local implementation, such as preservation of a unique habitat or a loss caused by a major oil spill. The issue of whose benefits should be measured has been described by Smith (1993, p. 21) as "probably more important to the value attributed to environmental amenities as assets than any changes that might arise from refining our estimates of per unit values." Empirical studies of goods with large passive-use value components (Loomis 2000; Sutherland and Walsh 1985) have verified that only a small percentage of the aggregate economic benefit accrues to individuals in the immediate jurisdiction where the good is located. Unfortunately, no simple rule of thumb exists for defining the study population when measuring passive-use values. Specification of the study population requires thoughtful consideration of who will benefit from provision of the good and who might realistically be asked or required to pay for the good.

3.2.1.2 Sample Frame

The sample frame is the list from which sample units (i.e., the individuals who will be asked to complete the survey) are chosen. Ideally the sample frame perfectly matches the study population, but that is rarely the case. Strictly speaking, if the sample frame does not perfectly match the study population, generalization from the survey sample should be made only to the sample frame, not to the study population. In practice, such generalizations are made when the sample frame and study population are well, but not perfectly, matched. When the sample frame does not perfectly match the study population, coverage error occurs because not all members of the population have a known nonzero chance of being asked to complete the survey (Figs. 3.2 and 3.3).

"Undercoverage" is the part of the study population that is not included in the sample frame. For example, a survey of property owners, such as those with properties in the neighborhood where the Wi-Fi service would be provided in the previous example, would miss renters. Undercoverage will also occur if the survey is administered in a manner that excludes some individuals in the population. For example, a survey with an email letter of invitation would exclude individuals who

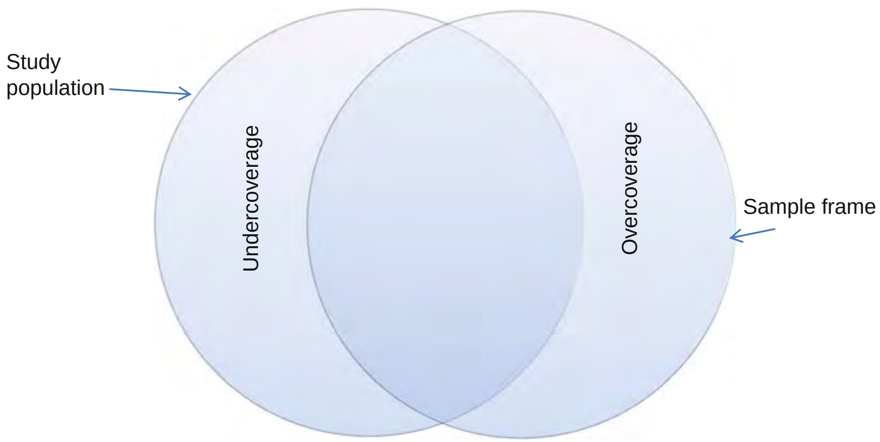

Fig. 3.3 Study population and sample frame

do not have an email address. Likewise, overcoverage occurs if the sample frame includes some individuals who are not in the study population, such as individuals who recently moved out of the study area or are deceased. Figure 3.3 illustrates the relationship between the study population and the sample frame.

Finding a single list of individuals or households in the study population with the requisite contact information might not always be possible. It is common for a sample frame to be developed from multiple sources, such as voter registration lists and tax assessor databases. For the Wi-Fi service case mentioned, if the researcher wants to measure the benefits to all residents in the neighborhood, she might consider a sample frame such as a property tax assessor database. Because this sample frame would fail to include individuals who are renters, she might augment the property tax assessor database with voter registration information. However, some residents might not be registered to vote, and the tax assessor database could include investors/landlords who are not residents. In such cases, purchasing a list of residents from a survey research firm might be a good option. Although coverage of the study population using such lists might not be perfect, it is often very good.

3.2.1.3 Drawing the Sample

After a sample frame has been developed, a sample can be chosen. There are two issues related to choosing a sample: how to choose sample units and how many sample units to choose. As mentioned previously, with probability sampling every unit in the study population has a known nonzero probability of being chosen; it is this fact that allows inferences to be made from the sample to the broader study population (or sample frame if the frame does not adequately represent the study population).

The most straightforward probability sampling technique is simple random sampling. With a simple random sample, every sample unit has an equal probability of being chosen, and it is not necessary to weight the final data (unless the data need to be weighted to correct for sample nonresponse).

However, many situations call for a more involved sampling technique than simple random sampling. For example, if the study population is geographically dispersed, a simple random sample can result in sample units that are also geographically dispersed. Geographic dispersion could be problematic for a survey that requires an interviewer to travel to the respondent to administer the questionnaire. An alternative in this case is cluster sampling, which can be used to minimize travel costs between on-site interviews. The general idea is that respondents are chosen in groups or clusters where the clusters are similar to each other, but the individuals within the clusters are representative of the population as a whole. For example, a simple cluster sample could define blocks in a city as clusters. First, blocks are selected and then potential survey participants within each of the blocks are chosen. Cluster sampling can be quite complicated, so consulting a survey statistician about sampling procedures is a good idea.

A third approach to random sampling, called “stratified random sampling,” divides the sample frame into nonoverlapping subpopulations (strata) based on some measure available for the entire sample frame. Stratified sampling differs from cluster sampling in that the strata differ with respect to the measure (e.g., income, education) used to stratify the sample but the individuals within a stratum are similar to each other.

Within each stratum, a variety of methods can be used to select the sample units. Stratified random sampling is primarily used in three situations. First, it can be used to ensure adequate sample size within strata for separate analysis. This is important when individual strata are of particular interest. For example, consider a nonmarket valuation study with the primary objective of estimating mean willingness to pay for improved access to a recreational area and a secondary goal of comparing the mean willingness to pay between urban and rural populations. In this situation, a simple random sample might not provide an adequate number of rural respondents to allow for detection of a statistical difference. The sample could be stratified into rural and urban strata, with the rural stratum more heavily sampled to ensure an adequate number of respondents for analysis. The second situation in which stratified sampling is used is when the variance on a measure of interest is not equal among the strata. In the rural-urban example above, there could be higher variance of willingness to pay for the urban population relative to the rural population. In this case, more of the sample can be drawn from the stratum with the larger variance in order to increase the overall sample efficiency. The third situation in which a stratified sample might be used is when the cost of obtaining a survey response differs by strata. An example of this would be implementation of an online survey where possible, with a phone survey for individuals who do not have access to the Internet. Survey responses from the phone stratum will be more costly than those from the online survey stratum.

Beyond stratified and cluster sampling, there are many kinds of complex multistage random sampling approaches; see Groves et al. (2009) for a good explanation of these.

In addition to deciding how to choose sample units from a sample frame, the researcher must decide how many units to choose. Two important considerations in choosing the sample size are sampling error and statistical power. Sampling error (Fig. 3.2) is the deviation that arises because we observe statistics (e.g., mean willingness to pay or average income) for the sample rather than for the entire sample frame. Because the value of the population statistic is not known (if it were known, it would not be necessary to do the survey), the sampling error is approximated using a probabilistic approach that is a function of the sizes of the sample and the associated population. For small study populations, a relatively large proportion of the population is needed in the sample to maintain a given level of sampling error. For example, for a study population of size 1,000, approximately 200-300 observations (completed surveys) are required for ±5% sampling error. The sensitivity of the sampling error to sample size decreases as the population increases in size. Whether the size of the study population is 10,000 or 100 million, a sample size of approximately 380 is needed to obtain ±5% sampling error, which is not much more than the sample size needed for the previous example with a study population of 1,000 (see Dillman et al., 2009, p. 57, for a table of completed sample sizes for various population sizes and different margins of error).

The second consideration when choosing the sample size is statistical power. After the survey is complete, the researcher will often conduct statistical analyses in which a null hypothesis (H_0) and an alternative hypothesis (H_1) are specified. Statistical theory identifies two types of error that can occur when conducting hypothesis tests. Type 1 error occurs if H_0 is rejected when it is true. Type II error occurs if H_0 is not rejected when H_1 is true. The power function $(P(\theta))$ is defined as the probability of rejecting the null hypothesis when it is false (thus, it is 1 minus the probability of a type II error):

$$P(\theta) = 1 - \beta(\theta),$$

where θ is the parameter of interest and $\beta(\theta)$ is the probability of type II error. In other words, the power of a test is low when the probability of type II error is high. The details of estimating a power function can be found in most introductory statistics textbooks. In addition, Mitchell and Carson (1989, Appendix C) provided a nice exposition on estimating the power function for testing the difference between two means. One key point is that as the overall sample size increases, the power function increases. If statistical testing involves subsamples of the survey data, the number of observations in each of the subsamples must be considered. Of course the completed sample size is usually much smaller than the initial sample, so the initial sample should be selected with consideration of the expected response rate to provide an adequate number of final observations.

Finally, the sample size is also related to the survey mode (Sect. 3.2.3). With a fixed budget, choosing a more expensive mode of administration, such as an interviewer-administered survey, implies a smaller sample size than is possible with a relatively less expensive mode, such as a mail survey.

3.2.1.4 Nonprobability Sampling

There are situations in which inference to a broader population is not important. As described in Chap. 10, economic experiments often use nonprobability sampling to recruit participants. Likewise, online panel surveys (Sect. 3.2.3.1) typically use nonprobability samples. The three commonly used nonprobability sampling techniques are convenience, quota, and purposive samples. An example of a convenience sample would be recruiting students from a class to participate in a survey. In this situation, the survey respondents are recruited because it is easy, not because they have particular characteristics related to the research. On the other hand, with a purposive sample, individuals are intentionally recruited because they have characteristics related to the study that could be difficult to find through screening general populations. For example, choice experiments (Chap. 5) are often used to value the attributes of new food products. If a researcher would like to administer a choice experiment survey about lematoes, a fruit that is a cross of a tomato and a lemon, a purposive sample might include individuals in the grocery store who purchase tomatoes. Quota sampling is an approach where the individuals are recruited up to a predetermined number of respondents with particular characteristics. For example, it could be determined that the researcher wants 100 Nordic skiers and 100 snowmobilers to complete a survey on winter recreation in a state park.

There are many permutations to each of the three nonprobability sampling approaches described here. Strictly speaking, generalization beyond a nonprobability sample should not be done. However, as a practical matter, nonprobability sampling has become more common as response rates to probability samples have been declining. Furthermore, adjustments are sometimes made to data from nonprobability samples post hoc in an effort to draw inference to a population beyond the sample.[1] One limitation of post hoc adjustments is that the survey population measures for which data are available might not be strongly related to the nonmarket values of interest. For example, adjusting a sample based on U.S. Census data measures of education and income to make inference to the population from a sample about mean willingness-to-pay values for open space might not be appropriate if willingness to pay for open space is not related to income or education. It could be the case that willingness to pay for open space is strongly correlated with individuals' perspectives on the appropriateness of government-provided public goods. However, data are not likely available for the study population on such attitudes. The bottom line is that the researcher needs to be very clear about the nature of the sample and the appropriateness of drawing inferences from it.

3.2.2 Designing a Questionnaire

The previous section described approaches to sampling or rules for identifying potential survey respondents. Decisions about sampling and decisions about survey design are intertwined because both are affected by the same constraints (e.g., time, budget, available labor). For the sake of exposition, the survey design process is described before the mode of administration. In reality, many decisions related to questionnaire design and survey administration are made simultaneously. Regardless of how a survey is administered, the design process for the questionnaire is largely the same.

The design of the survey materials, including the questionnaire, recruitment materials, reminders, and any other items developed to communicate with potential participants, can affect whether a respondent completes and returns the

[1]Post hoc adjustments are weights applied to the data so that the sample has a distribution on key variables (usually demographic variables) that is similar to the population of interest. Groves et al. (2009) described approaches for weighting data. Post hoc adjustments are also made to data collected using probability sampling to correct for nonresponse error.

questionnaire. Fortunately, the researcher has control over the appearance of self-administered questionnaires, the tone and professionalism of interviewers, and the description of why an individual should complete the survey. The information in Sects. 3.2.2.1-3.2.2.4 highlight information relevant to all survey modes of administration described in subsequent sections. For more details on how to design and administer a survey, consult the text by Dillman et al. (2009), which is considered the survey research bible.

3.2.2.1 Identify Desired Measures

The first step in developing the questionnaire is to identify what it is that the researcher would like to measure in the survey. For example, the researcher may want to consider what data are needed to estimate the nonmarket valuation model, and test hypotheses of interest. The researcher will usually have a good sense of the goals of a study but could struggle with identifying the specific measures needed to address those study goals. After the researcher has identified the desired measures, he or she will need to develop questions to quantify the desired measures. For example, a researcher designing a contingent-valuation survey to measure willingness to pay for improved fishing conditions might also want to measure respondents' angling experiences. However, simply asking the question, "Are you an experienced angler?" will not provide useful information if some anglers consider skill as a measure of experience while others consider the number of times fishing as a measure of experience. The researcher might want to develop questions about the specific dimensions of angling experience, such as the number of years the respondent has been fishing, how often the respondent goes fishing, and the type of gear the respondent uses when fishing. It is often useful to review studies on similar topics to facilitate development of appropriate questions to capture desired measures.

3.2.2.2 The Questionnaire

When designing the questionnaire, the researcher wants to minimize the measurement error (Fig. 3.2) that arises if the responses to a survey question do not accurately reflect the underlying desired measure. Poor design of the survey questions causes measurement error. The questionnaire should be clearly written using common vocabulary. Focus groups and other qualitative approaches described in Chapter 8 of Groves et al. (2009) can be used to investigate issues such as appropriate language for the respondent population and baseline knowledge of the survey topic. How questions are phrased will vary with the mode of administration. Interviewer-administered questionnaires often pose questions in the spoken language, while self-administered surveys are usually written to be grammatically correct.

Contingent-valuation and choice-experiment surveys (Chaps. 4 and 5, respectively) often include a detailed description of the good or the program being valued. The researcher should make sure the description is accurate and consistent with scientific knowledge. However, it is essential that the vocabulary be appropriate for the survey respondents. For example, if the questionnaire includes a description of a program to eradicate white pine blister rust from high elevation ecosystems, the researcher cannot assume that the respondents know what white pine blister rust or high elevation ecosystems are. Such concepts need to be defined in the description of the program.

Three important issues related to the design of questions are discussed below. It might also be useful to read books specifically on survey design, such as Marsden and Wright (2010), Groves et al. (2009), Dillman et al. (2009).

One consideration related to question design is whether to ask the question using an open- or closed-ended format. Open-ended questions do not include response categories. For example, the question "What do you like most about visiting Yellowstone National Park?" suggests that the respondent is to answer the question in his own words. Responses to open-ended questions can be categorized subsequently, but doing so is subjective and time consuming. Nonmarket valuation surveys usually feature closed-ended questions, which are less burdensome to respondents than open-ended ones, and which are more readily incorporated into statistical analyses. Developing response categories for closed-ended questions takes careful consideration of the range of possible answers. In developing response categories, it can help to use open-ended questions in focus groups or one-on-one interviews (see Sect. 3.2.2.4).

Another issue is confusing question design. One type of confusing survey question is the double-barreled question. An example is asking respondents to indicate their level of agreement with the statement, "I am concerned about the environmental effects and cost of suppressing wildfires." A respondent who is concerned about the environmental effects of wildfires but not the cost of suppressing them will have a difficult time responding to this statement.

Survey respondents could also misinterpret attitude statements that switch between positive and negative as the following statements do:

- National parks should be easily accessible by roads.
- National parks should not include designated wilderness areas.
- National parks should be managed to preserve native plant species.
- Motorized vehicle should not be allowed in national parks.

Respondents who are asked whether they agree or disagree with these statements might not notice that the statements switch between positive and negative. While it is important that such statements not appear to be biased toward a particular perspective, the statements should all be stated in a consistent manner.

In addition, the response categories should be consistent with the question asked. For example, the question, "Did you go fishing in the last 12 months?" should have "yes" and "no" as response categories. However, it is not uncommon to see this type of question followed with response categories such as “frequently,” “occasionally,” and “never”; one would want to avoid this type of inconsistency between the question and the response categories. Further, if the reference time frame changes between questions, it should be pointed out to the survey respondent. For example, if one set of questions in a fishing survey asks about expenditures in the last 12 months and another set of questions refers to expenditures on the most recent trip, the change in reference should be explicitly noted.

How to elicit recall data is another question design issue. Frequency questions, such as those about the number of doctor visits in the last 12 months or the number of visits to a particular recreational area, are often posed in nonmarket surveys. Frequency questions can be difficult for some respondents. Schwarz (1990) summarized the cognitive processes that respondents go through when responding to behavioral frequency questions:

> First, respondents need to understand what the question refers to, and which behavior they are supposed to report. Second, they have to recall or reconstruct relevant instances of this behavior from memory. Third, if the question specifies a reference period, they must determine if these instances occurred during this reference period or not. Similarly, if the question refers to their “usual” behavior, respondents have to determine if the recalled or reconstructed instances are reasonably representative or if they reflect a deviation from their usual behavior. Fourth, as an alternative to recalling or reconstructing instances of the behavior, respondents may rely on their general knowledge, or other salient information that may bear on their task, to infer an answer. Finally, respondents have to provide their report to the researcher. (p. 99)

Respondents might be better able to recall frequency of events if they are provided with cues that make the questions more specific. For example, instead of asking how many times a person went fishing, asking the number of times a person went fishing for specific species could reduce the cognitive burden on the respondent. Likewise when asking about expenditures for travel to a particular recreation destination, it is helpful to ask about particular expenditures (e.g., fuel, food, equipment) rather than total trip expenditures. Given the heavy dependence of some nonmarket valuation techniques—such as the travel cost method—on recall data, the researcher needs to be sensitive to potential data-collection problems and to implement techniques to mitigate such problems. Chapter 6 has more specific suggestions for eliciting trip recall information.

3.2.2.3 Question Order and Format

After developing the questions, the order and format of the questions need consideration. Again, decisions about question order and format are closely related to whether the questionnaire is interview-administered or self-administered. Sudman and Bradburn (1982) suggested starting a questionnaire with easy, nonthreatening questions. Because contingent-valuation and choice-experiment questionnaires can require substantial descriptions of the goods or attributes of the goods, some practitioners break up these large descriptions by interspersing relevant questions about the provided information. Dillman et al. (2009) suggested putting objectionable questions, such as the respondent's income, at the end of the questionnaire because respondents are more likely to complete the entire questionnaire if they have already spent five or 10 min working on it. Asking all demographic questions at the end of the questionnaire is common practice.

It is important to realize that the order of the questions can influence responses. This phenomenon could be of particular concern for stated preference surveys that ask the respondent to value multiple goods. While the researcher can tell the survey respondent to value each good independently, individuals might not be willing or able to do so. Indeed there is much evidence of order effects associated with contingent-valuation questions (Powe and Bateman 2003). However, Clark and Friesen (2008) also found order effects associated with actual purchase decisions of familiar private goods and concluded that order effects could be associated with general preferences rather than exclusive to stated preferences. Such effects are difficult to mitigate, so it is often recommended that researchers randomize the order of questions that are susceptible to order effects.

3.2.2.4 Testing and Refining the Questionnaire

The final version of a questionnaire often bears little resemblance to the first draft. Numerous revisions can be required before arriving at a well-designed questionnaire, one capable of engaging the respondents and providing meaningful data. Qualitative methods such as focus groups and one-on-one interviews are commonly used to refine and develop the questionnaire. Because nonmarket valuation surveys can involve topics with which respondents are not familiar, understanding the respondents' baseline level of knowledge is essential to writing the questions at the appropriate level. Qualitative methods are also used to understand how respondents think and talk about a particular topic. The vocabulary of a highly trained researcher and a layperson can be quite different. The questionnaire should avoid jargon and technical language that is not familiar to respondents.

After the questionnaire has been drafted, it should be reviewed by experienced nonmarket valuation practitioners to assure that the appropriate measures are included and the design is conducive to a good response rate as well as to producing quality data. In addition, the questionnaire and other survey materials should be tested or reviewed by potential respondents using qualitative research methods. While such methods are generally considered helpful for developing and refining survey materials, there is no agreement about standard procedures. The most common qualitative approaches used in survey development—focus groups and one-on-one interviews are described below.

A focus group is a moderated meeting with a small group of individuals to discuss the survey topic as well as drafts of the questionnaire and other survey materials. The goal is to develop survey materials and identify potential problems. The discussion is led by a moderator who has an agenda that identifies the topics to be covered within the limit of the meeting time (usually two hours or less).

However, the moderator should allow for some unstructured conversation among the participants. The balance between covering agenda items and allowing a free flow of conversation can pose a challenge to the moderator.

Another challenge is dealing with the diverse personalities of the focus group participants. A participant with a strong personality can dominate the focus group if the moderator is not able to manage that participant and draw out comments from the more reserved participants. The researcher should carefully consider how the small-group dynamic can influence what issues are raised and how important participants really think those issues are. For example, participants are sometimes susceptible to jumping on the bandwagon regarding issues that they would not have considered otherwise.

Some project budgets will allow for focus groups conducted at professional focus group facilities by an experienced moderator. However, much can also be learned using less expensive focus groups conducted in a more informal setting such as a university classroom or local library. The important attributes of a successful focus group include development of an appropriate agenda, a somewhat diverse group of participants, and a moderator who is adequately trained to obtain the desired information from the participants. Focus groups are usually recorded (audio and/or video) so the moderator can concentrate on the group discussion without being concerned with taking notes. Afterward, the recordings are reviewed and summarized. Additional details about how to conduct a focus group can be found in Greenbaum (2000).

Another qualitative method used to develop and refine nonmarket survey materials is the one-on-one interview, also called a cognitive interview (Groves et al. 2009). A one-on-one interview includes an interviewer and a potential survey participant. Unlike focus groups, the intimate context of a one-on-one interview removes the possibility of a participant being influenced by anyone other than the interviewer. These interviews usually take one of two forms: either the survey participant is asked to complete the questionnaire and review accompanying materials, such as a cover letter, while verbalizing all of his or her thoughts, or the interviewer asks questions from a structured list and probes respondents about their answers. In the latter case, after the respondent has answered the survey questions, the interviewer might ask about motivations for responses, interpretation of a question, or assumptions made when answering the questions. Enough interviews should be conducted to allow for a sense of whether issues are particular to one individual or are more widespread.

3.2.3 Survey Mode

Tradeoffs encountered in choosing the best mode for administering the survey include survey administration costs, time constraints, sample coverage, and context issues. Given the rapid changes in communication technology, it is best to have the survey administration plan reviewed by an experienced survey researcher because there could be approaches not considered or unforeseen pitfalls of a mode under consideration.

In general, surveys can be administered in two ways: a respondent can complete a questionnaire (self-administered), or an interviewer can orally administer the questionnaire and record the responses (interviewer-administered). Of course there are hybrid options as well. For example, questions can be asked orally and responses made in written form or entered on a keypad. The next two sections summarize key considerations when choosing between a self-administered survey and an interviewer-administered survey.

3.2.3.1 Self-Administered Surveys

A self-administered survey can be carried out using paper or an electronic device. Paper surveys are often mailed to potential respondents, but they can also be handed out at a recreation site or other location where the respondents have gathered. Electronic surveys (e-surveys) can be completed online, as an attachment to or embedded in an email, or on a device provided by the survey sponsor. For example, tablet computers can be used to complete questionnaires on-site. Self-administered surveys usually require fewer resources to implement than interviewer-administered surveys and, in turn, are often less expensive. The procedures for implementing a self-administered survey are also less complicated than those of interviewer-administered survey. Furthermore, interviewer effects are avoided, and the questionnaire can be completed at the respondent's pace rather than the interviewer's.

The main drawback of self-administered surveys is that there are several potential sources of error. Nonresponse errors (Fig. 3.2) can arise if respondents examine the survey materials to see if they are interested in the topic and then decide not to participate. This can result in important differences between respondents and nonrespondents. In addition, completion of a self-administered questionnaire requires that respondents be competent readers of the language.

An additional potential drawback of a self-administered paper questionnaire is the loss of control over the order in which questions are answered. Respondents to a self-administered paper questionnaire can change their responses to earlier questions based on information presented later in the survey. It is not clear if this would be desirable or undesirable; however, it is an uncontrollable factor. Control over order can be maintained in a self-administered e-survey. Finally, if the survey is self-administered via the postal service, there can be a long time lag between when the survey packet is sent out and when the questionnaire is returned.

E-surveys are gaining in popularity and offer many advantages over both self-administered paper surveys and interviewer-administered surveys. The cost per completed e-survey can be lower than self-administered paper surveys or interviewer-administered surveys. In addition, the responses to an e-survey are entered electronically by the survey respondents, saving time and money. The downside to an e-survey is that it can be completed on different types of devices, such as laptop computer, a tablet computer, or a smart phone, and formatting for all the different devices can be tricky. Also, setting up an e-survey can require expertise in computer programming. However, there are programs available that can be used to set up a relatively straightforward e-survey. Lindhjem and Navrud (2011) reviewed stated-preference nonmarket studies that compare e-surveys to other modes of administration and also reported on their own contingent-valuation study that compared results of an in-person, interview-administered survey to an e-survey administered online. They found that the share of “don’t knows,” zeros, and protest responses to the willingness-to-pay question was very similar between modes, and equality of mean willingness-to-pay values could not be rejected.

Self-administered e-surveys are often administered to online panels made up of individuals who agree to complete multiple e-surveys. Numerous approaches are used to recruit panel participants. While some panelists could continue to participate indefinitely, attrition is an ongoing issue, so new panel members might need to be continuously recruited. Panel participants can refuse to complete any individual survey and are often compensated for their participation. Some, but not most, online panels are recruited using probability-based methods. In such cases, some recruits might not have Internet access, so it is provided by the agency recruiting the online panel.

When a researcher contracts with a company to implement a survey online, the company will usually agree to provide a specified number of completed surveys. The costs of panel online surveys are usually lower than those of an interviewer-administered survey and might be less than a mail survey. If the researcher wants to generalize the results of the survey to a broader population, the panel company might guarantee that respondent demographics match the population of interest, or the survey data could be weighted to match the demographic characteristics of the study population. However, using weights with a nonprobability sample might not be sufficient for making inference to a broader population. As discussed in Sect. 3.2.1.4, such adjustments can only be made using measures that are observable for the broader population (i.e., demographic characteristics). If the observable characteristics of the broader population are found to be unrelated to the nonmarket values in a statistically significant manner, weighting the panel data to draw inference about the nonmarket values of a population broader than the panel participants is called into question. In addition, a nonprobability online survey sample does not provide a framework, such as that provided in Fig. 3.2, for identifying sources of error (Tourangeau et al. 2013).

The advantage of a panel survey to the researcher is the guarantee of a given number of completed surveys from respondents with demographics that match a broader population. Further, the surveys are usually administered quickly, so the researcher has data in hand in a short period of time. However, not all panel companies are equivalent, so the researcher should investigate the methods of a particular company.

The American Association for Public Opinion Research provides recommendations regarding use of online panels (Baker et al. 2010). Its first recommendation is “Researchers should avoid nonprobability online panels when one of the research objectives is to accurately estimate population values." However, the association also concludes, "There are times when a nonprobability online panel is an appropriate choice. Not all research is intended to produce precise estimates of population values, and so there could be survey purposes and topics where the generally lower cost and unique properties of Web data collection are an acceptable alternative to traditional probability-based methods." Finally, it makes clear that users of online panels need to make sure they understand the nature of the particular panel: "Users of online panels should understand that there are significant differences in the composition and practices of individual panels that can affect survey results. Researchers should choose the panels they use carefully" (p. 714).

At the request of the U.S. Environmental Protection Agency, the Wyoming Survey and Analysis Center recently compared results obtained using an online panel with those obtained from mail and phone surveys. The sample for the mail and phone surveys was based on a random-digit dialing list with reverse address look-up purchased by the Wyoming Survey and Analysis Center from a vendor. Knowledge Networks, the company that administered the online panel, used a very similar sample for recruitment into the panel. The details of the sampling are described in Grandjean et al. (2009), Taylor et al. (2009). The authors found that response rate was highest for the mail survey (30%), followed by the phone survey (16%), then the online panel survey (4%). The mail and phone survey response rates were computed using a comprehensive formula identified by the American Association for Public Opinion Research (Smith 2015) that accounts for not only the individuals known to be eligible (i.e., they were successfully contacted and met the criteria for participation) but also for the individuals who were never successfully contacted in initial phone efforts. The bottom line is that comprehensive estimates of response rates are low for all three modes, but they are particularly low for the online panel. In the online panel survey, many potential respondents were lost because they either did not agree to participate in the panel initially or they left the panel prior to the survey being implemented. However, the completion rate (i.e., number of completed surveys per number of individuals identified as potential participants) is highest for the phone survey (91%), followed by the online survey (77%) and the mail survey (75%). As is evident in this study, completion rates can be quite different from response rates.

Grandjean et al. (2009) also found some statistically significant differences across modes in the demographic characteristics of the respondents. For example, the online panel participants were less likely to belong to an environmental organization than the phone or mail respondents. The measure of interest in this comparison study was willingness to pay for cleaner air in national parks. The willingness-to-pay estimate based on the phone survey was higher than the willingness-to-pay estimate from the mail or online panel surveys. The authors speculated that this result is due to social desirability bias (described in the following section) associated with the phone survey and concluded that the online panel survey provided a willingness-to-pay estimate as accurate as the well-designed mail survey. However, without a thorough investigation of potential nonresponse bias, it is not clear how the willingness-to-pay estimates from the survey respondents compare to those of the broader population.

3.2.3.2 Interviewer-Administered Surveys

Interviewer-administered surveys have traditionally been conducted in-person or over the telephone. However, current technology also allows for an interviewer in a remote location to administer a survey via technology that transmits video, voice, or text over the Internet. Interviewers can administer surveys in respondents' homes, in convenient locations such as shopping malls or on the street, or in places where the participants of targeted activities tend to congregate. For example, recreation use surveys are often administered at the site of the recreational activity (e.g., a hiking trailhead, boat landing, or campsite).

The primary advantages of interviewer-administered surveys come from the control the interviewer has while administering the survey. The interviewer can control who completes the survey as well as the order of questions. In addition, the interviewer can provide clarification if the survey is complex. If a survey is administered in-person, a system can be developed for choosing participants that avoids the need for an initial sample frame with contact information. Such an approach is often used in countries where complete lists of individuals who reside in an area are unavailable. Finally, if an interviewer administers a survey, the respondent does not need to be literate.

An important downside of an interviewer-administered survey is that it tends to be more expensive than a self-administered survey due to interviewer costs. The interviewers need to be paid for training, conducting the interviews, and travel costs if the surveys are conducted in-person. Another major concern with interviewer-administered surveys is that the presence of an interviewer can affect survey responses (Singer and Presser 1989). In particular, social desirability bias—where the presence of the interviewer results in a more socially desirable response (e.g., a higher willingness to pay or higher number of trips to a recreation site reported) than would be elicited in a self-administered survey—is a concern for nonmarket valuation surveys.

3.2.4 Administering the Survey

If the questionnaire and other survey materials are finalized, the plan for drawing the sample completed, and the survey mode selected, the survey is almost ready for administration. The next two sections cover approaches for contacting potential survey respondents and for conducting a pilot test of the survey procedures before final implementation.

3.2.4.1 Contacting Potential Survey Respondents

Recruitment of potential survey respondents need not employ the same mode as the administration of the final survey. Potential survey respondents can be contacted via the postal service, email, social media sites, the telephone, or in-person (e.g., at their residences or a public location, such as a recreation site or a mall). The information in the sample, such as telephone numbers and mailing addresses, will dictate the options for contacting potential survey respondents. Sometimes the invitation to complete the survey and the subsequent administration of the survey are done together. For example, if the sample includes address information, a survey packet can be mailed with a cover letter inviting potential respondents to complete an enclosed paper questionnaire. As mentioned earlier, mixed-mode surveys are currently the norm. For example, a letter could be sent through the mail to invite a potential respondent to complete an online survey.

3.2.4.2 Pilot Survey

A pilot survey is a small-scale test of the draft survey materials and implementation procedures. The pilot survey serves several purposes. First, it allows for a test of the survey sample. Some samples can be problematic, but the researcher might not realize it until the survey is in the field and great expense has already been incurred. For example, I was involved in the design of a mail survey in which a nonprobability sample was developed using a list of registered voters.[2] Even though the list was current, during the pilot study we found that many of the addresses were not valid, and a substantial number of the surveys were undeliverable. In the end, a completely different sample was used for the final survey.

The pilot survey also provides information about the anticipated response rate, the number of follow-ups that will be needed, and the expected cost for the final survey. Third, the pilot study allows for a test of the survey implementation procedures. For contingent-valuation studies, the pilot study also offers an opportunity to learn about the distribution of willingness to pay. For example, if the researcher plans to use a dichotomous-choice contingent-valuation question in the final survey, an open-ended question can be used in the pilot study to guide assignment of the offer amounts for the final survey (see Chap. 4 for details of contingent-valuation surveys).

[2]The study was a split sample test of different decision rules in a contingent-valuation question. Because we did not need to draw inference to a larger population about the magnitude of the estimated contingent values, a nonprobability sample was used.

3.2.4.3 Survey Implementation

After the survey materials have been designed and the implementation procedures tested, the survey is ready to go into the field. Regardless of the mode of administration, final survey implementation tends to be hectic. A mail survey requires a substantial effort. The questionnaire and cover letters need to be printed, the postage and envelopes need to be purchased, and the survey packets need to be assembled for mailing. If a survey is going to be interviewer-administered, the interviewers must be trained in the survey administration procedures.

One approach to survey implementation is to offer multiple modes for completing the survey. For example, respondents can be offered a choice of completing an online or a mail questionnaire. The motivation for a mixed-mode survey is that some respondents might be more likely to respond to their preferred mode, resulting in a higher response rate than with a single-mode survey. As part of an annually administered survey, Olson et al. (2012) asked respondents in 2008 about their preferred mode for future surveys. They found that for Web-only and phone-only mode preferences, the response rate was significantly higher for the individuals who received their preferred mode compared to those who did not receive their preferred mode in the 2009 survey. Likewise, results of a meta-analysis suggest that mail surveys with a simultaneous Web option have lower response rates than a mail-only option (Medway and Fulton 2012). The bottom line is that intuition that mixed modes will increase response rates is not backed up with strong evidence.

A database should be set up with the information from the sample that allows for tracking respondents and nonrespondents. In addition to tracking whether an individual has responded, it is beneficial to track when an individual responded and when follow-up contacts were made. This information can be used for planning future surveys, and it allows for testing for differences between early and late respondents.

Because most nonmarket valuation surveys elicit sensitive information such as annual household income, it is important to assure respondents that their responses will be confidential and their names will not be associated with their responses. The standard procedure is to assign a unique identification number to each sample unit in the initial sample to track whether or not the person has responded. A unique identification code should also be used with online surveys to make sure that only those invited to complete the survey do so and that no one completes the survey more than once. The identification numbers, not the respondents' names or addresses, are included in the final data set. Likewise, survey interviewers have an ethical responsibility to keep all responses confidential.

Surveys conducted through universities usually require informed consent, which involves informing respondents that their participation is voluntary. The specific information provided to respondents about the terms of their participation and the procedures for getting permission from a university to do a survey vary among universities. The funding organization for the survey should be knowledgeable about both the procedures for obtaining permission to conduct a survey and the procedures for obtaining consent from the survey respondents.

3.2.4.4 Survey Responses

Ideally, all contacted individuals would complete the survey; however, only a portion of the contacted individuals will actually respond. The response rate to a survey is the number of completed surveys divided by the number of individuals who were invited to complete the survey. As long as the individuals who respond to the survey are not different from those who do not respond to the survey in a manner that is relevant to the study objectives, a response rate of less than 100% is of no concern, and nonresponse error is nil. However, a synthesis of the research on nonresponse (Groves 2006) suggests that nonresponse error is present in many surveys.

There is confusion in the literature about how nonresponse error is related to the response rate. It is often suggested that a higher response rate will decrease nonresponse error. For example, the guidelines for statistical surveys sponsored by U.S. government agencies suggest that an analysis of nonresponse error should be conducted for response rates lower than 80% (U.S. Office of Management and Budget 2006). Groves and Peytcheva (2008) conducted a meta-analysis of the effect of response rates on nonresponse bias and found that the response rate by itself was not a good predictor of nonresponse bias. In other words, nonresponse bias was found in some surveys with relatively high response rates and not found in some surveys with relatively low response rates. They did find that general population surveys, surveys sponsored by government agencies, and surveys of study populations that did not have previous knowledge about the study sponsor were associated with higher nonresponse bias. There is no hard rule about what is an acceptable response rate; rather, a reasonable response rate is one that provides enough observations for the desired analyses and a group of respondents with characteristics similar to the population to which inferences will be made. If possible, an analysis of nonresponse bias should be conducted.

3.2.5 Completed Surveys

If the surveys were completed electronically, a useable dataset will be provided automatically. In other cases, several additional steps are required. The first step is to develop a set of rules, called a codebook, for data entry. Then the data are entered and verified. Finally, data analyses can be conducted.

3.2.5.1 Codebook

Coding refers to the process of translating the text responses into numerical responses. For example, a question that elicits a yes or no response could be coded so that a “1” represents a yes response and a “0” represents a no response. A codebook is developed for the entire questionnaire; it specifies the variable names

Have you ever visited Yellowstone National Park? (Circle one number)

1 Yes

2 No

Fig. 3.4 Sample survey question

for each survey question. One option to make the coding process easier is to include the numerical codes next to the response categories on the questionnaire, as shown Fig. 3.4. If, however, the responses are recorded by filling in a bubble or putting an "X" in a box, a numerical code might need to be written on the survey prior to data entry. Even when responses have corresponding numbers, some judgment can be involved in coding the data if the respondent did not follow the directions. Therefore, in addition to simple rules, such as 1 = yes and 0 = no, the codebook specifies numerical codes related to skipping a question. Weisberg et al. (1996) described standard coding practices.

Coding open-ended questions is more complicated. One option is to transcribe entire responses to open-ended questions. Analyzing such data would require use of a qualitative method such as content analysis. Another approach is for the researcher to develop a workable number of response categories and code each open-ended response into the appropriate category.

3.2.5.2 Data Entry

If surveys are not administered electronically, the data must be entered into an electronic database. Improvements in database management software have made data entry fairly easy. The software can be set up to notify the data entry person when a number that is out of the relevant range has been entered or to skip questions that the respondent was supposed to skip but did not. These innovations minimize data entry errors. However, even when using a data entry software package, the entered data should be checked. One option is to randomly compare surveys responses and the electronic data file. Another option is to enter the data twice. During the second entry of the data, the computer software signals if an entry does not match the original data entry.

After the data have been entered, frequencies should be run on all variables and checked to make sure the response values are within the appropriate range for each question and that skip patterns have been properly followed. Since a unique identification number is entered in the dataset and written on the actual questionnaire, errors in the data can be checked by going back to the original surveys. This step is referred to as data cleaning.

If the survey was conducted with the intent of generalizing to a broader population, the socioeconomic variables from the survey respondents should be compared to those of the broader population. If the survey respondents differ from the broader population to which survey results are to be generalized, sample weights can be applied. Quite a few articles describe how to weight respondent populations (see Whitehead et al. 1993, 1994; Dalecki et al. 1993; Mattsson and Li 1994; Whitehead 1991).

3.2.5.3 Documentation

Regardless of the study objectives, all studies should include documentation of the procedures used to collect the data as well as summary statistics on all measures elicited in the survey. Proper documentation will make the data more accessible. Chapter 11 describes the benefit transfer approach, where results from an original study are used in a different context. For an original study to be amenable to benefit transfer, the original study must be appropriately documented. Complete documentation will also assist reviewers of the original study. Loomis and Rosenberger (2006) provided suggestions for reporting of study details to improve the quality of benefit transfer efforts. The best way to document data collection procedures is to do it as the study is being conducted. The entire study—from the initial study plan to the final summary statistics—should be documented in one location. Table 3.1 outlines the information about the study that should be documented. In addition, data should be archived. Oftentimes the funding institution provides guidelines on how data should be archived.

3.3 Secondary Data

Nonmarket values can often be estimated using secondary (i.e., existing) data. Secondary data might or might not have been collected with the original intent of estimating nonmarket values. Although use of secondary data can save substantial expense and effort, secondary data should only be used if the researcher is confident about the quality. In the U.S. many large national surveys, such as the U.S. Decennial Census, are well-designed and collect data that could be used for nonmarket valuation. The U.S. Census Bureau administers many other useful demographic and economic surveys, such as the National Survey of Fishing, Hunting, and Wildlife-Associated Recreation. Most of the data collected by the Census Bureau can be downloaded from its website (www.census.gov). Likewise, Eurostat, the statistical office of the European Union, provides statistics and data from European countries (www.epp.eurostat.ec.europa.eu).

An incredible amount of data from cell phones, the Internet, Global Positioning Systems, and other sources is continually being created. These so-called "big data" sources are a relatively new development and have not been used extensively in nonmarket valuation. However, it seems likely that these new data sources will soon provide important valuation opportunities. One can imagine the travel cost models described in Chap. 6 using data provided by a fishing app (application

Table 3.1 Guidelines for documenting a study

	Item to document	Information to include
Step 1	Study plan	Date Initial budget Study proposal
Step 2	Focus groups/one-on-one interviews	Dates Location Costs (amount paid to participants, food, facility rental, etc.) Focus group moderator/interviewer Number of participants (do not include names or addresses of participants) Handouts Agenda Video and/or audio recordings Interview or focus group summaries
Step 3	Sample	Description of study population Description of sample frame Procedures for selecting sample units Cost of sample Number of sample units Data included with sample
Step 4	Final survey	Text of letter or script used to invite potential respondents Questionnaires Dates of all survey contacts Number of respondents to each wave of contact
Step 5	Data	Codebook Final sample size Frequencies (including missing values) for each measure in the survey Means/medians for continuous measures Data file in format useable for data analysis software

software) or from information about visitation to rock climbing areas from GPS units. Cukier and Mayer-Schoenberger (2013) provided an overview of what big data sources are and how they are changing the way we approach data. Of course, while the allure of huge samples and lots of information make big data an exciting proposition for nonmarket valuation, the researcher should weigh these benefits against the structure and control available with an original survey.

Another secondary data source often used in nonmarket valuation is confidential data. For example, the information used to estimate damage costs described in Chap. 8 often includes hospital and/or urgent care admissions data. Use of such data requires strict adherence to procedures developed to maintain confidentiality. The provider of such data might require that no results be reported for small subsamples that could potentially allow for identification of individuals affected by the disease or health symptom of interest. The application process for obtaining confidential data usually specifies how the data can be stored, used, and reported.

A consideration when using any kind of secondary data is whether the data contain the requisite variables for estimating the desired nonmarket value. It is not only an issue of having the variables or not—the variables must be in the appropriate units. If, for example, one wants to estimate a travel cost model using some expenditures that are in annual terms and other expenditures that are reported on a per-trip basis, the expenditures must be converted so that both are on either a per-trip basis or an annual basis. Another consideration is whether the data are representative of the appropriate population. The key sampling issues that need to be considered when collecting primary data, such as defining the relevant study population and making sure the final data are representative of that population, should also be considered when assessing the appropriateness of secondary data.

Geospatial data are often used to augment nonmarket valuation data. One such source is Landsat data (http://landsat.gsfc.nasa.gov/data/), which uses satellite imagery of the earth to monitor changes in land use, water quality, encroachment of invasive species, and much more. These data are maintained by the U.S. Geological Survey and are available for free (http://landsat.usgs.gov/). Researchers can use these data in nonmarket valuation models as an objective measure of the change in the quality or quantity of an environmental amenity (or disamenity). When augmenting a nonmarket valuation model with geospatial data, the researcher must ensure that the scales of the different data sources are similar.

3.4 Conclusion

Many individuals come to nonmarket valuation with a background in economics that facilitates understanding of the associated theory and estimation methods. However, few newcomers to nonmarket valuation arrive with survey research experience. The goal of this chapter was to provide guidance on how to collect primary data or assess the quality of secondary data to support estimation of valid nonmarket values.

A survey effort starts with identifying the relevant study population. This straightforward-sounding task involves many considerations, such as who will benefit from provision of the nonmarket good and who will pay for it. The researcher must also consider how potential respondents will be contacted and how the survey will be administered. Rapid changes in communication technology are opening new options for administering surveys while also complicating many aspects of traditional survey research methods.

Perhaps the most difficult task in developing a nonmarket valuation survey is moving from notions of what the researcher would like to measure in a survey to the articulation of actual survey questions. This chapter provided general guidelines to help nonmarket valuation newcomers through the survey process.

Now that the general economic perspective on nonmarket valuation has been introduced in Chap. 1, the conceptual framework presented in Chap. 2, and nonmarket data collection methodology described in this chapter, the reader is prepared for the next eight chapters that describe specific nonmarket valuation techniques. The last chapter in this book focuses on reliability and validity of nonmarket valuation. As that chapter discusses, high quality data alone will not guarantee the reliability or validity of nonmarket values; however, quality data are a prerequisite for reliable and valid measures.

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